Sensor Technology for Ice Hockey and Skating

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***Abstract*—Sensor technology that is unobtrusively integrated into the clothing and equipment of an athlete can support the training of sport activities and monitor the athlete's progress. In this paper, we propose two wearable systems that support ice hockey players in the training of skating and shooting. These assistants measure the motions of players and compare them with reference executions of the same activities by professional players. A third system that we introduce monitors the player's activities during a hockey game and creates a match report for objective performance measurement. For each of the three proposed applications, we present a prototype setup that we evaluate with amateur and professional players. The main findings are i) that with a skate-worn motion sensor and user-dependent training, eight skating motions can be spotted with an accuracy above 90%, ii) that stick-integrated sensors enable the measurement of relevant shot features, which differentiate professional from amateur athletes, and iii) that it is possible to spot important ice hockey activities in the signals of body-worn motion sensors worn during a game.**

I. Introduction

Ice hockey is a fast and intense sport played worldwide by 1.7 million licensed and many more hobby players in non-official leagues [1]. Relevant skills for hockey players include ice skating, the handling of the hockey stick and physical plays such as body checking and fist fights [2]. To learn how to shoot while moving at speeds of up to 30 km/h requires a high level of coordination and much practice. Even though players wear protectors for all body parts, they are at a high risk of injuries, in particular concussions due to hits against the head [3].

For the study of injuries, to improve hockey and skating equipment, and as a complementary tool to professional coaching, wearable sensor setups have been proposed: Previous work includes the development of a sensor-based system for analyzing power strokes, the main accelerating motion in skating [4], and helmet-integrated inertial sensors for monitoring the high impacts during hockey playing. These impacts may result in concussions and spine-related injuries [5]. With the FWD PowerShot[1] device by Quattriuum, a system for measuring the timing of shooting motions in hockey is commercially available. Furthermore, literature presents a variety of research prototypes and commercial systems for related team sports, in particular for American football [6], field hockey [7] and baseball [8].

[1]http://www.quattriuum.com/en/powershot

In this paper, we discuss potential applications for state-of-the-art wearables in skating and hockey, and we present prototype systems and their evaluation with amateur and professional players. We thereby focus on the following aspects:

- **Skating:** We present the *SkateTracker* system for monitoring skating and recognizing relevant events, in particular *power strokes*, *breaking*, *turns*, *jumps* and *curves*. The system provides basic statistics about each movement, e.g., the timing and intensity of *power strokes*. Given the similar motion patterns, the same system is also tested in inline skating, which is a popular sport by itself: According to [9], $13 - 19$ million people go inline skating regularly in the US alone. To the best of our knowledge, SkateTracker is the first *step counter* optimized for the typical skating motion patterns.

- **Shooting:** Previous research on shooting techniques in ice hockey used high-speed cameras and motion capturing [10], [11]. However, camera-based systems are not applicable in regular training situations because of the associated installation effort. The above-mentioned, stick-integrated FWD PowerShot[1] system can track stick orientation and timing, but it does not consider other characteristics of a shot execution such as the hand positions on the stick. In this work, we equipped the *Sensorized Hockey Stick* with motion sensors, strain gauges and pressure-sensitive potentiometers. With the sensorized hockey stick, players can measure the stick's motion in space, its flexion, and the hand positions while playing. The stick wirelessly transmits data to a smartphone for provision of feedback that helps to improve the shooting technique. The system automatically differentiates between shot types, and recognizes the user's skill level based on his handling of the device.

- **Gameplay:** In soccer and other sports, sensor-based game monitoring already proved to provide relevant information about an athlete's performance [12]. For ice hockey, no such system exists as of now. We present the wearable sensor setup *GameTracker* for the identification of relevant hockey game events, such as *hits*, *shots*, *time in motion*, etc. This wearable monitoring system can provide statistics about a game without the need for external surveillance.

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II. SKATING

A. Problem Introduction

Ice and inline skaters use similar accelerating motions: During the *power stroke*, one foot is aligned with the skating direction, while the other foot pushes the skater forward. For fast skating, the switching time between left and right foot should be as short as possible, as there is no accelerating force acting on the athlete while one foot is in the air [13]. The most frequently applied breaking technique in ice skating is the so-called *hockey stop*, for which the skater sets both feet orthogonally to the skating direction. As an alternative, inline skaters can decelerate by pressing a heel brake mounted behind the back wheel of one skate against the ground.

Aside from accelerating and breaking, a number of other skating movements are particularly relevant in ice hockey, since players continuously need to position and orient themselves with respect to their opponents and the puck. Motions that hockey players perform include *narrow curves* with both feet parallel and on the ground, *wide curves* with intermediate strokes, *forward-backward* and *backward-forward turns*, *backward skating*, and *jumps*, e.g., to avoid an obstacle.

B. Method

In this work, we focus on the recognition and analysis of the above-listed motions with Inertial Measurement Units (IMUs) that are clipped to the shoelaces of the skate. Our *SkateTracker* system fulfills two purposes: First of all, it detects skating motions in the IMU acceleration and rotation velocity data. Secondly, SkateTracker derives features for each recognized skating action, such as timing characteristics and maximal accelerations for *power strokes*, and the foot's orientation at a specific point in time. As of now, all calculations are performed offline in Matlab. However, in future, a smartphone application could provide immediate feedback and act as an improved wearable step counter for skating, and as a training assistant that indicates when a person performs movements in a non-ideal way (e.g., too long time between strokes).

We implemented two standard machine learning approaches for the detection of skating motions: *feature-based classification* and *template matching*. The reviews in [14] and [15] provide detailed introductions for both techniques. In feature-based classification, we use a sliding window to segment the data into windows of size WS, with an overlap $SS = WS/8$. For each window, SkateTracker extracts a total of $N_f = 80$ statistical (mean, standard deviation, min, max, number of zero crossings, etc.) and frequency features (fast Fourier transform coefficients) from the raw data. Feature selection based on mutual information reduces this set to the N_f^r most relevant features. Next, SkateTracker trains a classifier for these features, and applies the model to detect skating motions at runtime. Our alternative approach for motion detection, *template matching*, follows the steps proposed in [16]. We use k-means clustering to fuse the multi-dimensional accelerometer and gyroscope input data into a one-dimensional sequence of symbols (string). Next, SkateTracker applies the Warping LCSS algorithm [17] to calculate the similarities between a training template and the test data. If the similarity exceeds the threshold t_{LCSS}, SkateTracker recognizes the corresponding skating action.

For optimal performance, we subdivided the problem of detecting skating motions into three categories: First of all, the *short activity* category, which includes *jumps*, *turns* and *breaking*; second, *longer activities*, which include *narrow* and *step curves* and *backward skating*; third, *power strokes*, which we consider as an equivalent to steps in pedometers. SkateTracker fuses the system outputs and assures that it recognizes at most one skating motion at a time. In this fusion, long activities have priority, followed by short activities and power strokes. This for example means that the system does not count stroke motions during *curves* as regular *power strokes*.

C. Evaluation

We recorded two datasets for parameter optimization and evaluation of the motion detection algorithm: A first dataset D_1 with a single amateur inline skater performing a circuit ten times. Each round included at least one repetition of the following motions: *Power strokes*, *forward* and *backward turns*, *backward skating*, *jumps*, *narrow* and *wide curves* and *breaking*. In the second dataset D_2, six skaters of various skill levels performed the same circuit on ice between once and three times (in total ten recordings, of which one had to be excluded due to sensor issues). In both cases, the experiment participants had an EXLs3 IMUs[2] mounted on their right skate. For ground truth, we recorded videos with a GoPro HD2 camera and labeled them with the annotation tool ANVIL[3].

All the analyses in this section are event-based, i.e., if a detected skating action overlaps in time with a ground-truth label of the same type, this recognition is counted as one true positive observation. For each ground truth label, only a single overlapping action is counted as a true observation, all others are considered as insertion errors. We performed an initial parameter optimization by leave-one-run-out cross-validation applied to the inline skating dataset D_1. The goal of this optimization was to maximize the F1 score, which equally weighs precision and recall. We tested a set of state-of-the-art classification algorithms and sweeped the window size WS and classifier settings. Overall, random forest classification with at least 25 subtrees performs best for short and long activities. In D_1, the optimal window size for short activities is $WS = 0.6$ s, and in long activities $WS = 1.0$ s. For $N_f^r \geq 20$ features, the classifier output does not improve much further.

For power strokes with their regular motion pattern, template matching outperformed all feature-based classification algorithms, achieving an F1 score of 0.94 in detecting strokes, compared to 0.88 with Random Forest. With an optimal alphabet size $k = 250$ and an LCSS rejection threshold $t_{LCSS} = 0.25$, the algorithm on average correctly recognizes 113 power strokes in D_1, with 2 insertion and 9 deletion errors. This result is achieved with two randomly selected training templates, excluding the initial and last strokes of all skating sequences. Most of the deletions are related to starting power strokes, which produce different motion patterns compared to strokes during regular skating. Figure 1a) depicts the overall confusion matrix, combining short and long activities with power strokes. The overall F1 score of SkateTracker in D_1 is 0.92 (precision 0.95, recall 0.90).

[2] http://www.exelmicroel.com/products_medical_exls3.html
[3] http://www.anvil-software.org/

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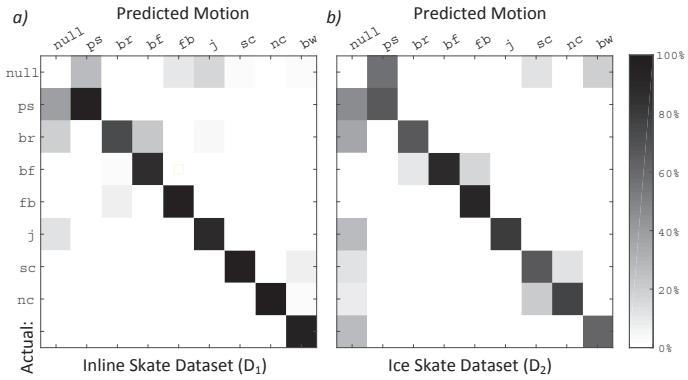


Fig. 1. SkateTracker confusion matrices in the two datasets for *power strokes (ps)*, *breaking (br)*, *backward-forward turns (bf)*, *forward-backward turns (fb)*, *jumps (j)*, *step curves (sc)*, *narrow curves (nc)* and *backward skating (bw)* with optimal parameter settings for each category. The color indicates the percentage of actions per combination of recognition and ground truth motion. Results are averaged for 10 repeated algorithm executions with identical parameters. In D_2, there are more deletion errors due to the larger variation with which participants executed the motions. In particular, some participants performed the curves and turns in anti-clockwise rather than clockwise direction. The template-matching-based power stroke detection worked much less reliable in D_2, with an average of 35 true positives, 19 deletion errors and 13 insertions. Again, this is most likely due to the large differences in speed and intensity among the experiment participants.

In the ice skating dataset D_2, we re-applied the same parameters, except that we adapted the threshold t_{LCSS} in the power stroke detection to be 0.2. This adaptation was necessary because of the by far larger variance in the execution of power strokes in this dataset (given that players of various skill levels participated). Again, we applied a leave-one-run-out cross validation to study the performance of the skating motion detector in D_2. Since some players performed turns and breaking in clockwise direction and others in anti-clockwise direction, the training data for curves, turns and breaking had to be split into subactivities, which decreased the overall performance as less training data per motion class was available. Figure 1b) depicts the confusion matrix for this user-independent analysis. The overall F1 score decreases to 0.65 in D_2. Most of the errors are related to power strokes (F1 score for power strokes only: 0.53). For future, we recommend user-dependent training of the stroke detection, or adaptive algorithms that replace default stroke templates with actual data of the user at runtime.

III. Shooting

A. Problem Introduction

The stick is the hockey player's tool for passing, dribbling and shooting. Next to skating, stick handling is the most important component of practice in hockey. In this section, we consider sensor modalities that could be used for analyzing stick handling, with a focus on the two main shooting motions: the wrist shot and the slap shot [18]. For a wrist shot, the player starts with the stick's blade touching the puck, accelerating it in a forward motion. This shot allows for precise targeting, but the puck does not reach as high speeds as in a well-executed slap shot. For slap shots, the player first winds up the stick, and then hits the ice about $0.2-0.3$ m in front of the puck with maximal force. The stick flexes and once it touches the puck, a spring-like unloading will maximize the impulse. Amateur players usually shoot pucks at speeds of around 120 km/h, while professionals reach speeds of up to 190 km/h.

B. Method

We here present the Sensorized Hockey Stick that we equipped with a set of sensors and Bluetooth Smart (former Bluetooth Low Energy) communication to a smartphone. The processing and communication unit of the stick is a Bonsai Sensing Platform[4] that we placed together with an IMU (Bosch Sensortech BNO055) and AAA batteries in a custom design case. This case fits into the cavity at the upper end of the stick, where it is protected from impacts. In addition, we placed two pressure-sensitive linear potentiometers (Spectra Symbol Thinpot, 0.5 m) on the outside of the stick and covered them with tape. With these sensors, we can read out the hand positions of the athlete on the stick. Four strain gauges (Micro Measurements, CEA-06-062UW-350) glued to the lower part of the stick measure its flexion. We amplify the strain signals before transmission to the sensing platform with an AD8231 amplifier. The strain gauges are glued with two-component adhesive to the stick, and a layer of Polydimethylsiloxan (PDMS) protects them from impact and moisture. Figure 2 depicts the stick prototype. The IMU and hand position data are sampled at a rate of 100 Hz, and stick flexion information at 400 Hz. Figure 3 depicts the raw data extracted with the stick for a slap shot executed by a professional player. As we did not measure the relationship between the strain gauge measurements and the actual bending radius of the stick, we normalized the flexion values to be in $[-1, 1]$. Hand positions are measured as distance from the upper end of the stick.

The analysis of recorded shots is done in two steps: First, the systems detects that a shot took place and classifies it as either a wrist or slap shot. The rule-based classification algorithm that we implemented recognizes peaks in the stick deflection to identify the moment of shots. The algorithm then considers the stick orientation immediately before the deflection peak to decide whether this was a slap or a wrist shot. If the player wound up, the shot was a slap shot and vice-versa. After recognizing and classifying the shot, the system extracts a set of characteristic features for each shot. This includes the maximum, minimum and mean rotation velocities of the stick in all three axes during the shot, acceleration and timing characteristics (e.g., duration between maximum wind up and ice contact), the maximum flexion of the stick, the hand positions, etc. Aside from displaying these features to athletes and coaches, we used them and further standard statistical features extracted from the sensor data (as in [14]) for training a Support Vector Machine (SVM) classifier that assesses a player's skill level based on his shot execution.

C. Evaluation

We built two sensorized hockey sticks for players that shoot on the left and right side, and we recorded data with 11 amateur and 8 professional players from the Swiss National League team EHC Olten. Amateur players were volunteers that we recruited in public ice skating rinks. All the amateur participants were familiar with the basic shooting technique,

[4]http://www.bonsai-systems.com/

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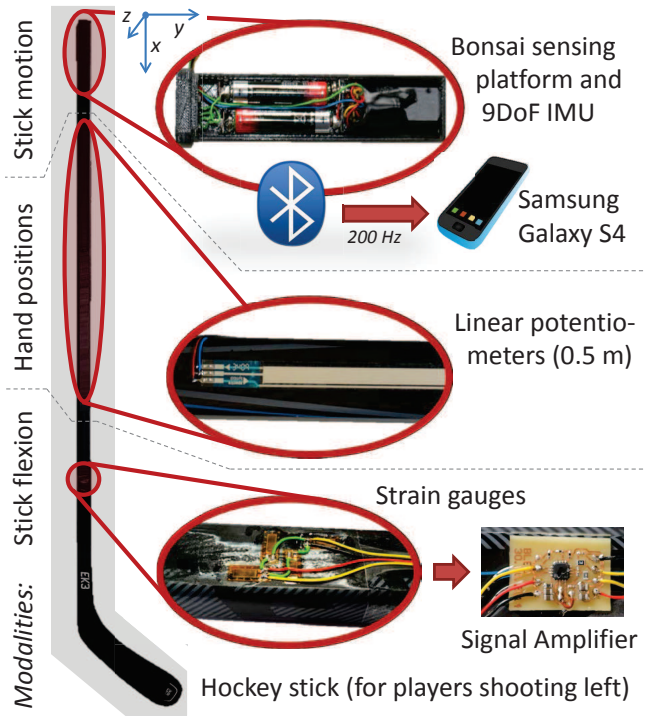


Fig. 2. This figure shows the *sensorized hockey stick* and its sensor components, which are usually not visible as they are either placed inside the stick, or covered with tape.

but they did not practice with a professional coach or play in an official league at the time of the recording. Unfortunately, issues with loose battery contacts repeatedly resulted in data loss at the time of high impacts, in particular during slap shots. For this reason, we had to pre-select shots with little or no data loss prior to the analysis, and ended up with a subset of 64 shots by amateur players (43 wrist, 21 slap) and 36 shots by professionals (26 wrist, 10 slap) for further analysis. At the time of writing, we solved the issue with the loose battery contact through modifications to the battery case. The rule-based classifier for detecting these shots and their type worked with an accuracy of 100%.

Figure 4 depicts a set of selected shot features for amateur and professional players. We used these and further features to train a classifier that distinguishes the shots of beginners and professional athletes. In a leave-one-shot-out cross validation, this algorithm achieved an accuracy of 88.5% for wrist shots and 92.9% in slap shots. As amateur players may at times by chance perform a shot well, a classification based on a single shot is not reliable for skill assessment. However, when classifying a player based on a majority vote in five slap shots, the classification accuracy improves to 99.7%. We applied Sequential Forward Selection (SFS, [19]) to identify the features with the highest significance for distinguishing amateur from professional players. According to an evaluation in the full dataset, the most relevant features for slap shots are the minimal rotation velocity in shooting direction (around y-axis), the duration of the pre-swing phase, and the maximum flexion of the stick (see Figure 4). Also among the top-five features for slap-shot-based skill assessment is the lowest position of the lower hand during the shot. For wrist shots, the highest selected features are again related to the lower hand position,

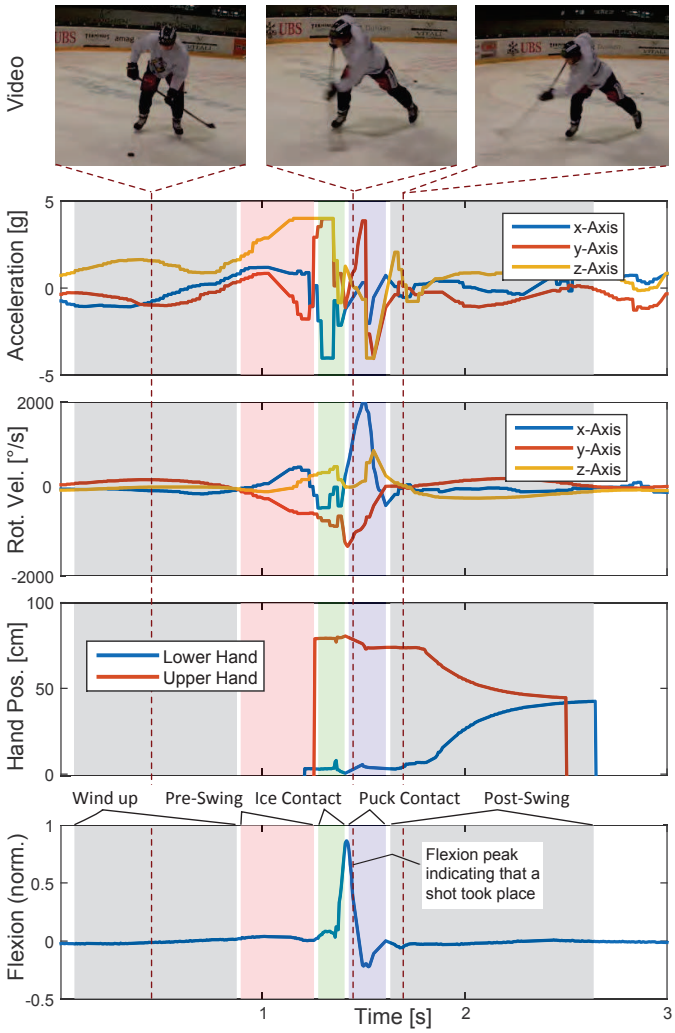


Fig. 3. This plot shows the raw hockey stick signals for an expert slap shot, with the different shot phases indicated. During the *wind up* phase, the player rotates the stick around the y-axis in positive direction. Then, in the *pre-swing* phase, he rotates the stick back to the ice. The stick flexes during *ice contact* and then unloads after *puck contact*. The puck's impulse also leads to accelerations and rotations around the stick's main axis that are well visible in the sensor data. In the *post-swing* phase, the athlete stops the stick's rotation around the y-axis and returns to the initial position. The accelerometer saturates at 4 g, and the hand position sensors only measure contacts when the hand's pressure is sufficiently high. As the data shows, this is usually the case immediately before *ice contact* and until the end of the *post-swing*. In the *post-swing* phase the player's lower hand moves up and in the case of the presented shot, it touches at one point both, the upper and lower position sensors, which affects the position estimates towards the end of the shot.

the minimal and maximal rotation velocities around the y- and x-axes, and the summed up rotation around the x-axis. These features highlight the main difference between amateur and professional shooting, and their selection by SFS corresponds well with what professional coaches propose to practice for improved shooting (short swing phase, low positioning of the lower hand, maximal stick flexion, etc. [10]).

Finally, we implemented a smartphone app that performs the feature extraction at runtime for provision of immediate feedback. With this app, the user can load a reference shot and display the above-listed and further relevant features for comparison with his last executed shot. The app highlights

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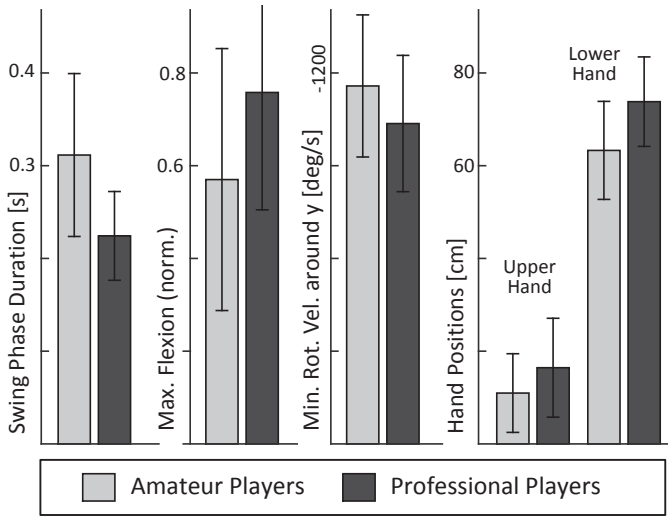


Fig. 4. These are averaged values for some of the extracted features from the slap shot dataset with amateur and professional players. As expected, the pre-swing phase is shorter for professional players, and the maximal flexion of the stick higher. Experts also place their hands lower from the upper end of the stick.



Fig. 5. This figure shows an experiment participant with the sensors attached to his equipment. The smartphone for logging sensor data was mounted on the inside of the trousers.

differences between the user's execution of a shot and the way professionals perform the same movement. Therefore, it could be used as a training assistant with real-time feedback, and to monitor the long-term progress in the execution of hockey movements.

IV. Gameplay

A. Problem Introduction

In professional hockey, monitoring the activities of players during matches is an objective way of evaluating their performances. Assistants track the actions of each player and count shots, goals, hits, time on ice, etc. However, this effort is only affordable for elite teams in the American, Russian and Scandinavian leagues. In other leagues, youth hockey and amateur matches, players are mainly evaluated according to subjective observations by their coaches.

We here propose to use wearable sensors that can be integrated into the protections of the players to monitor the following game events: *Wrist shots*, *slap shots*, *hits*, *time in motion* and *breaks*. These are also the events that the major hockey leagues typically monitor for each player. By synchronization of sensor data with regular game statistics (e.g., the time of goals), a game tracker can decide whether a shot resulted in a goal or not, or whether a player was on the ice when a goal happened. In addition, we identify *turns* and *backward skating* from the sensor data, which reveals when a player switches between offense and defense.

B. Method

As in the skating motion classification, we use wearable IMUs to identify gameplay events. Here, we attach the sensors to the gloves and the upper-body protector of the athlete, where they are at a lower risk of getting damaged in real game situations. We use feature-based classification to spot gameplay events, and extract statistical and frequency features in windows with size WS and overlap $SS = WS/8$. Unlike in skate tracking, we here combine the data of three IMUs, which triples the number of features to consider.

The gameplay tracker trains two individual Random Forest classifiers: A first one for the recognition of short motion events (*slap* and *wrist shots*, *hits*, *turns*), and a second one for recognizing periods of *time on ice* versus *breaks*. A third classifier that detects whether a player is in *puck possession* or not performed very poorly in our datasets and we had to discard the approach.

C. Evaluation

In a preliminary experiment, we used EXLs3 IMUs that we attached to the player's upper body and both his wrists. These IMUs sampled acceleration and rotation velocity at 100 Hz and streamed the data to a Samsung Galaxy SII phone that the players wore with them. Figure 5 depicts the sensor setup. As some of our experiment participants were right shooters, we mirrored their motion sensor data before feature extraction and aligned it with the motion data of players shooting left.

With this setup, we recorded the motions of 4 players of the professional team EHC Olten while performing rounds on a circuit that included *wrist* and *slap shots*, *hits*, *forward* and *backward skating*, a *short break* on the bench, and *time with* and *without the puck*. We also recorded videos for ground-truth annotation. In total, the players performed 10 rounds, and we applied a leave-one-round-out cross-validation to study the system's performance on this dataset D_3. Figure 6 depicts the confusion matrix for the hockey classifier with optimized parameters. The overall F1 score for the event detection with the optimal window size $WS = 0.6$ s in D_3 is 0.70 (precision 0.73, recall 67). The classifier accuracy in recognizing whether

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a player was on the ice or not is 86% for a window-based evaluation. Differences in the way players perform hockey motions, in particular the direction of turns, are the main reason for classification errors. In a next step, we will record more training data and then apply the classifiers to recordings during actual hockey games.

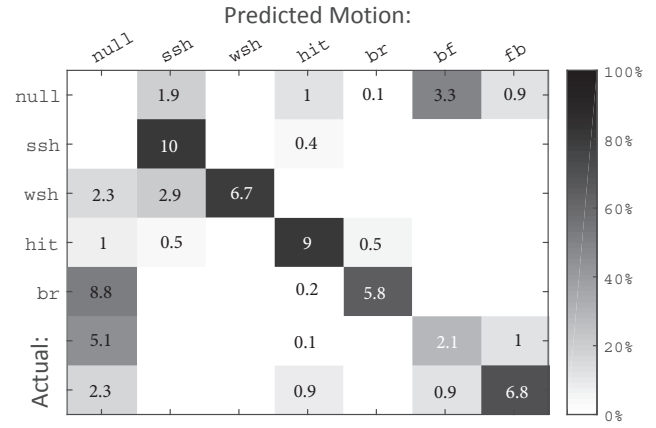


Fig. 6. Gameplay tracker confusion matrix in D_3 for *slap shots (ssh)*, *wrist shots (wsh)*, *hits*, *braking (br)*, *backward-forward turns (bf)*, and *forward-backward turns (fb)* with optimal parameter settings. Results are averaged for 10 repeated leave-one-run-out cross-validations with identical parameters. While shooting and hitting is recognized with high accuracy, differences in the direction of turns affected the classifier performance in the remaining classes. The numbers in the confusion matrix reflect the averaged number of occurrance of each combination between ground-truth and predicted activity.

V. Conclusion and Outlook

In this paper we presented ways of instrumenting skaters and hockey players with sensors for applications in training and game monitoring. Our first contribution is the SkateTracker system for the detection of skating motions that reaches an F1 score of 0.92 for user-dependent training. In a next step, this SkateTracker could be extended to provide feedback on the execution of each movement, and suggest ways for improving the technique. The second contribution of this work is the sensorized hockey stick, which provides shot execution data from stick-integrated sensors. We showed that it is possible to classify the shots of amateur and professional players with features extracted from the stick's sensor data, reaching an accuracy of 92.9% after a single shot. In future, we will collect a library of good shots, executed by professional players of different size and weight. Then, players that want to practice can select an ideal shot for them and see how they can improve their technique. Finally, this paper proposes a prototype for gameplay tracking and the spotting of a player's actions during a hockey game, reaching an F1 score of 0.7. In future, we will integrate the gameplay tracking sensors into the protections of players. Such an integrated system will be less obtrusive than our prototype and applicable in actual games.

Overall, our studies with prototype systems confirm that wearable sensors are able to capture important characteristics of the motions involved in hockey playing and skating. They can provide objective data to support coaches in giving feedback to players, and players themselves for training with real-time feedback when no coach is around. The technologies and methods that we propose in this paper are also applicable for similar purposes in other sports.

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